



VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Exploratory Data Analysis of the Iris Dataset

Course Code	MBAV 443M
Course Title	Statistical Modeling and AI-Driven Analytics
Assignment No.	Assignment 1
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Submission Date; 14 June 2026

1. Introduction

The Dry Bean Dataset is a key resource in machine learning for classification tasks, containing measurements of seven dry bean varieties: SEKER, BARBUNYA, BOMBAY, CALI, DERMASON, HOROZ, and SIRA. It includes numerical attributes related to the beans' geometric and shape traits, along with a categorical variable for bean classification. The assignment focuses on conducting Exploratory Data Analysis (EDA) to uncover insights, assess quality, detect anomalies, and examine variable relationships, setting the foundation for further analytical modeling.

2. Data Understanding and Quality Assessment

The Dry Bean Dataset began with 13,611 observations and 17 variables, which included 16 numerical features reflecting the beans' physical characteristics and one categorical feature for bean variety. A thorough data quality assessment was performed, checking data types, missing values, and duplicates. All variables were confirmed to be appropriately typed for analysis, and missing value checks revealed no absent data. However, 68 duplicate entries were found and eliminated using the `drop_duplicates()` function, resulting in a refined dataset of 13,543 observations. The data cleaning process enhanced the dataset's reliability for subsequent analyses, indicating it is clean and largely ready for exploratory analysis with minimal preprocessing required.

3. Methodology

The analysis was performed using Python within a Jupyter Notebook environment. The libraries utilized included Pandas, NumPy, Matplotlib, and Seaborn. The analytical workflow consisted of the following stages:

1. Dataset loading and inspection.
2. Verification of dataset dimensions and variable names.
3. Data type validation using `info()` and `dtypes`.
4. Missing value assessment using `isnull()` and `isna()`.
5. Duplicate record identification and removal.
6. Descriptive statistical analysis using measures such as mean, median, mode, standard deviation, variance, minimum, maximum, and quantiles.
7. Examination of categorical variables using `unique()`, `nunique()`, and `value counts()`.
8. Group-wise analysis using `groupby()` and aggregation functions.
9. Correlation analysis to investigate relationships among numerical variables.

10. Development of visualizations to explore distributions, detect outliers, and identify associations among variables.

Descriptive statistics were used to summarize central tendency and variability, while graphical methods facilitated the interpretation of patterns and relationships within the dataset.

4. Visualization

Various visualization techniques were applied to the Dry Bean Dataset. Histograms and kernel density plots assessed numerical variable distributions, while boxplots identified outliers and data spread. Violin plots provided insights into the distribution among bean varieties. Count plots showcased the frequency of the seven bean classes, and bar plots compared average measurements. Scatter plots explored relationships and clustering patterns, and correlation heatmaps visualized associations among numerical features, indicating potential overlap among some measurements.

5. Key Insights

The exploratory analysis of the Dry Bean Dataset revealed high data quality, with no missing values and the removal of 68 duplicates. An uneven distribution of bean varieties indicated class imbalance, impacting future classification models. Significant variability in physical characteristics was observed across seven bean classes, with correlation analysis showing strong positive relationships among numerical attributes. Overall, the analysis highlighted that these characteristics can effectively discriminate between bean varieties.

6. Conclusions

This exploratory analysis showcased the Dry Bean Dataset's structure, quality, and characteristics, revealing no missing values and few duplicates. Descriptive statistics and visualizations highlighted significant patterns among seven bean varieties and their physical attributes, suggesting that the features are suitable for further analysis. The study underscores the value of Exploratory Data Analysis (EDA) in understanding datasets and addressing potential issues prior to model development. Future work may focus on developing machine learning classification models to predict bean varieties based on geometric measurements and assessing their predictive effectiveness.

AI Usage Disclosure

Generative AI tools were used for coding assistance, conceptual clarification, and language refinement during the preparation of this assignment. All data analysis, validation of results, interpretation of findings, and conclusions remain the responsibility of the author.